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SkyeAv

Skye Lane Goetz

Applied AI/ML Engineer | Biomedical Data Scientist

Skills

Languages: Python, Elixir, SQL, Shell, LaTeX, Nim, Julia

AI/ML: PyTorch, LightGBM, Scikit-learn, LangGraph, MCP, Transformers (HuggingFace), PEFT, Diffusers, Ollama, LlamaCPP, WhisperCPP, Snorkel

Data & Storage: Polars, DuckDB, PostgreSQL, SQLite, ChromaDB, Spark, NumPy

Knowledge Graphs: Biolink, KGX, LinkML, Ontology Mapping, Graph Centrality

NLP: spaCy, NLTK, Sentence Transformers, BERT, TF-IDF, Text Classification

DevOps/Infrastructure: Nix/NixOS, Docker, FastAPI, Kubernetes, Mix (Elixir), Pytest

Domain: Multiomic Data, Shotgun Metagenomics, Polygenic Risk Scores, Organic Chemistry, DFT

Work Experience

Institute for Systems Biology (ISB) — **Seattle, WA**
Research Associate (Data Science) [Hood Lab] — *Nov 2024 - Present*

- Engineered MultiomicsKG, a 1M+ edge (QC in progress) multiomics knowledge graph integrating genomics, proteomics, and microbiome data across 52+ papers; a superset of published MicrobiomeKG [Tablassert]
- Refactored knowledge graph extraction from 800GB SQLite to 300GB DuckDB OLAP, reducing query time from ~10 minutes to 4–11 seconds (~100–1000x) using Polars lazy frames, Pydantic schemas, and compact CURIE encoding
- Built QualifiersMCP [GitHub], an MCP server for authoring and validating Tablassert table configs against Biolink qualifier ontology; deployed via Nix flake
- Co-mentored a master's student on a clinical trial classifier; expanded the pipeline with BERT sentence transformers, incremental PCA, and LightGBM, improving AUC from 0.63 to 0.90 across 3 datasets
- Developed a LangGraph-based AI agent for automated knowledge graph configuration from natural language, using a LoRA/P-tuning fine-tuned LLM via HuggingFace PEFT
- Developed consortium MCP tooling and a FastAPI + Neo4j query interface for LLM-accessible MultiomicsKG subgraph retrieval
- Published MicrobiomeKG [paper], a 115k-edge microbiome knowledge graph spanning 40 curated papers with ResNet-based edge scoring

Cal Poly — **San Luis Obispo, CA**
AI/Tech Lead – Parker, OpenAI-Sponsored Cal Poly Hackathon [GitHub] (~30% acceptance) — *2025*

- Led a LangGraph pipeline for course-material knowledge graph generation using Canvas API ingestion, LinkML ontology modeling, DuckDB storage, and LLM entity extraction from PDFs, PPTs, and Excel files

Zigler-Haynes Lab (CalPoly) — **San Luis Obispo, CA**
Student Researcher (Computational Subgroup) — *Sep 2024 - Nov 2024*

- Ran TD-DFT calculations on thiocarboxylate derivatives and built a Julia ORCA wrapper that parsed outputs 3x faster than prior lab scripts

Institute for Systems Biology (ISB) — **Seattle, WA**
Data Science Intern [Hood Lab] — *Aug 2024 - Sep 2024*

- Developed the Tablassert framework and co-authored the MicrobiomeKG preprint

Institute for Systems Biology (ISB) — **Seattle, WA**
Summer Intern/Trainee (NSF REU) [Hood-Price Lab] — *Jun 2024 - Aug 2024*

- Built a 40k-edge MicrobiomeKG prototype and presented it to the 200+ member CATRAX consortium

CalPoly Water-Energy Sustainability Training Team (WESTT) — **San Luis Obispo, CA**
Student Researcher (PHOS and CRIAS projects) — *Jan 2024 - Jun 2024*

- Scaled and Maintained Algal Cultures for Wastewater Treatment and Biofertilizer in Lab and Field Settings
- Monitored Culture Growth and Contamination through Microscopy, OD Measurements, and Mass Assays

Compass Group USA — **San Luis Obispo, CA**
Food Service Worker — *Jan 2024 - Jun 2024*

- Communicated with Team Members using Intermediate Spanish
- Occasionally Repaired Dishwashing and Kitchen Equipment

General Polymer Solutions (GPS Polymers) LLC **Upland, CA**
Customer Fulfillment and Logistics *Aug 2021 - Sep 2023*
- Streamlined Inventory Management with Improved Excel Formulation
- Coordinated Pickups with UPS, FedEx, and Other Couriers

US Soccer Federation **Upland, CA**
US Soccer Grassroots Referee (Grade 8) *Jun 2021 - Sep 2023*
- Worked with Associations and as a Private Contractor to Secure Work
- Dealt with Difficult Parents During High Stakes Games

Self Employed **Upland, CA**
Math Tutor *Jun 2023 - Sep 2023*
- Worked with Clients (Grades 3-5) to Improve Subject Performance
  - Turned Two C Students into A Students

Kessen Star Spinners **Upland, CA**
Founder, President, and CEO *Dec 2015 - Sep 2016*
- Founded and Ran a Fidget Spinner Company with 4 Employees
- Used FreeCAD to Customized Fidget Spinner Designs to Customer Specifications and for our 3D Printers
- Coordinated with Various Organizations to Sell Our Product
  - Made Over $2,000.00 in Revenue in One Night at a Local Little League Opening Day

Publications

Education

California State Polytechnic University (Cal Poly) **San Luis Obispo, CA**
BS Biochemistry *Feb 2024 - Present*
- GPA: 3.626/4.000
- Graduation Date: Spring Quarter 2026
- Consulted on Machine Learning Senior Projects for Computer Engineering Students
  - Improved AUC by 0.15
- Developed and Trialed Computational and Machine Learning Techniques for Biochemical Assays in Lab
  - Improved R^2 Coefficients by 0.08 on Average
- Trialed Modifications to Organic Syntheses in Lab
  - Speed Up Cited Procedure by 30 Minutes
- Consulted on Multinomial Classification Efforts in the Bioinformatics Lab

California State Polytechnic University, San Luis Obispo **San Luis Obispo, CA**
BS Agribusiness *Sep 2023 - Feb 2024*
- Completed 5 Months of Coursework Before Transitioning to Biochemistry
- Wrote a Comprehensive Report on the Economics of the California Almond Industry for the 2022 Growing Season

Upland High School **Upland, CA**
High School Diploma *Aug 2019 - Jun 2023*
- GPA: 4.000/4.000
- High School Soccer 4 Years
  - Team Captain 3 Years
  - Varsity Coaches Award
  - Varsity Most Improved Player Award
  - 2021-2022 Golden Boot
- Designed (with CAD) and 3D Printed Ceramic Stamps for the Ceramics Studio
- Wrote AP Physics Notes Programs for Class TI84s

Talks and Presentations

Conference and External.....

- **This Time on MultiomicsKG: Edge Scoring!** – NCATS Biomedical Data Translator Fall Relay (Bethesda,MD) Sep 2025

- **Gut Microbiome Signatures of Vegan, Vegetarian, and Omnivore Diets across 21,561 Individuals** – ISB Micro Journal Club (Seattle, WA) Aug 2025
- **Updates on the New MultiomicsKG** – NCATS Biomedical Data Translator Summer Relay (Bethesda, MD) Jun 2025
- **MicrobiomeKG: Extracting Relationships between the Microbiome and Host Health from Published Supplementary Materials** – Cal Poly Bailey College Research Symposium (San Luis Obispo, CA) May 2025
- **MultiomicsKG: Extracting Knowledge Assertions from Tabular Data in Supplementary Materials** – NCATS Biomedical Data Translator Spring Relay (Bethesda, MD) Feb 2025
- **MicrobiomeKG (v0.1): A Microbiome-Centric Knowledge Graph Facilitating Human Reasoning Across the Biomedical Domain** – ISB Summer Research Poster Session (Seattle, WA) Jun 2024

Group and Lab

- **This Time on MultiomicsKG: Edge Scores and Agentic Configurators** – ISB Hood Lab Meeting (Seattle, WA) Aug 2025
- **MicrobiomeKG: Connecting the Microbiome and Host Health through Knowledge Graphs** – ISB Thorsson-Shmulevich Lab Meeting (Seattle, WA) Jul 2025
- **MicrobiomeKG: Connecting the Microbiome and Host Health through Knowledge Graphs** – ISB Gibbons Lab Meeting (Seattle, WA) Jul 2025
- **Tablassert(3.0): Transforming Tabular Data Into Knowledge Graphs** – ISB Hood Lab Meeting (Seattle, WA) Feb 2025
- **MicrobiomeKG: Story of My Internship at ISB** – ISB Hood-Price Lab Meeting (Seattle, WA) Aug 2024

Projects

Tablassert — **NCATS Translator (at ISB)**
Table Mining Declarative Knowledge Graph Builder

MicrobiomeKG — **NCATS Translator (at ISB)**
A Knowledge Graph Connecting the Microbiome and Host Health

Table Config Agent — **NCATS Translator (at ISB)**
Agentic Natural Language Configuration for Tablassert-Based Knowledge Graphs

Clinical Trials Classifier — **NCATS Translator (at ISB)**
Binomical Classification for the AACT Clinical Trials Database

MultiomicsKG — **NCATS Translator (at ISB)**
A Superset of MicrobiomeKG Connecting Multiomic Factors with Host Health